Week 1 – Data Foundations & Cloud ETL

London Success Academy Joint Internship Programme
Data-SPARK Mentorship – Venkat Potamsetti

Introduction

Welcome to the London Success Academy Joint Internship Programme.

This internship introduces students to real-world data engineering workflows through structured weekly projects. The program is designed to simulate industry environments where data engineers must design pipelines, organise data, and support business analytics.

The mentorship is led by Venkat Potamsetti, creator of the Data-SPARK mentoring framework, which focuses on:

- Structured thinking
- Data pipeline design
- Cloud data engineering
- Documentation discipline
- Translating business problems into data workflows

Throughout the internship, students will work with realistic business scenarios involving property analytics and e-commerce data platforms. Each week increases in complexity to gradually build real industry capability.

Week 1 Scenario

Company: UrbanNest Property Analytics Ltd

UrbanNest is a London-based property analytics startup helping real estate agencies understand housing trends and property values.

UrbanNest recently received a dataset containing housing market data from several districts. The company wants to build a cloud-based data pipeline that allows analysts to query the data easily.

Unfortunately, the data engineering lead is on vacation, and you have been asked to step in and prepare the initial data pipeline.

Your task is to upload the dataset, catalog it, and allow analysts to run SQL queries.

Dataset Description

The dataset contains property information used to analyse housing values.

Column	Description
Rooms	Average number of rooms per property
Distance	Distance from employment hubs
Value	Median property value

This dataset allows analysts to study property price patterns and urban development trends.

Week 1 Learning Objectives

- Upload datasets to cloud storage
- Organise a data lake structure
- Automatically detect schema using AWS Glue
- Run simple ETL jobs
- Query data using SQL through Amazon Athena

Week 1 Tasks

Task 1 – Upload Dataset to AWS S3

Create a bucket: urban-nest-data-lake

Upload the dataset: house_prices.csv

Structure the data lake: s3://urban-nest-data-lake/raw/house_prices.csv

Task 2 – Create AWS Glue Crawler

Navigate to: AWS Glue → Crawlers

Create crawler named: urban-nest-house-crawler

Data source: S3 → urban-nest-data-lake/raw

Output database: urban_nest_db

Task 3 – Run the Crawler

- Detect the dataset schema
- Create a table in the AWS Glue Data Catalog
- Allow the dataset to be queried

Task 4 – Build ETL Job in AWS Glue Studio

- Reads data from the Glue Data Catalog
- Removes null values
- Saves cleaned data to: s3://urban-nest-data-lake/processed/
- Output format: Parquet

Task 5 – Query Using Amazon Athena

Example query:

```
SELECT AVG(value)
FROM urban_nest_db.house_prices;
```

Another query:

```
SELECT rooms, AVG(value)
FROM urban_nest_db.house_prices
GROUP BY rooms;
```

Week 1 Deliverables

- AWS Glue crawler screenshot
- Glue job workflow
- Athena query results
- Documentation of pipeline architecture